

JAVA PRACTICAL 3

1(a). Transpose of a 2D Array

```
import java.util.Scanner;

public class Transpose {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int r = sc.nextInt();

        System.out.print("Enter columns: ");
        int c = sc.nextInt();

        int a[][] = new int[r][c];

        System.out.println("Enter elements:");

        for (int i = 0; i < r; i++) {
            for (int j = 0; j < c; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Transpose:");

        for (int i = 0; i < c; i++) {
            for (int j = 0; j < r; j++) {
                System.out.print(a[j][i] + " ");
            }
            System.out.println();
        }
    }
}
```

1(b). Multiplication of Two 2D Arrays

```
import java.util.Scanner;

public class MatrixMultiplication {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows of first matrix: ");
        int r1 = sc.nextInt();

        System.out.print("Enter columns of first matrix: ");
        int c1 = sc.nextInt();

        System.out.print("Enter rows of second matrix: ");
        int r2 = sc.nextInt();
    }
}
```

```

        System.out.print("Enter columns of second matrix: ");
        int c2 = sc.nextInt();

        if (c1 != r2) {
            System.out.println("Matrix multiplication not possible");
            return;
        }

        int a[][] = new int[r1][c1];
        int b[][] = new int[r2][c2];
        int result[][] = new int[r1][c2];

        System.out.println("Enter first matrix:");

        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c1; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Enter second matrix:");

        for (int i = 0; i < r2; i++) {
            for (int j = 0; j < c2; j++) {
                b[i][j] = sc.nextInt();
            }
        }

        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                for (int k = 0; k < c1; k++) {
                    result[i][j] = result[i][j] + a[i][k] * b[k][j];
                }
            }
        }

        System.out.println("Result:");

        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                System.out.print(result[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

1(c). Product of Diagonal Elements and Column Elements

```

import java.util.Scanner;

public class MatrixProduct {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
    }
}

```

```

        System.out.print("Enter rows: ");
        int r = sc.nextInt();

        System.out.print("Enter columns: ");
        int c = sc.nextInt();

        int a[][] = new int[r][c];

        System.out.println("Enter elements:");

        for (int i = 0; i < r; i++) {
            for (int j = 0; j < c; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        int diagonalProduct = 1;

        for (int i = 0; i < r && i < c; i++) {
            diagonalProduct = diagonalProduct * a[i][i];
        }

        System.out.println("Product of diagonal elements = " +
        diagonalProduct);

        for (int j = 0; j < c; j++) {
            int product = 1;

            for (int i = 0; i < r; i++) {
                product = product * a[i][j];
            }

            System.out.println("Product of column " + (j + 1) + " = " +
        product);
        }
    }
}

```

1(d). Sort the Elements in 2D Array

```

import java.util.Scanner;
import java.util.Arrays;

public class SortArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int r = sc.nextInt();

        System.out.print("Enter columns: ");
        int c = sc.nextInt();

        int a[][] = new int[r][c];
    }
}

```

```

int b[] = new int[r * c];

System.out.println("Enter elements:");

int k = 0;

for (int i = 0; i < r; i++) {
    for (int j = 0; j < c; j++) {
        a[i][j] = sc.nextInt();
        b[k] = a[i][j];
        k++;
    }
}

Arrays.sort(b);

k = 0;

System.out.println("Sorted array:");

for (int i = 0; i < r; i++) {
    for (int j = 0; j < c; j++) {
        System.out.print(b[k] + " ");
        k++;
    }
    System.out.println();
}
}

```

1(e). Sum of All Elements Ending with 4

```

import java.util.Scanner;

public class SumEndingWith4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter rows: ");
        int r = sc.nextInt();

        System.out.print("Enter columns: ");
        int c = sc.nextInt();

        int a[][] = new int[r][c];
        int sum = 0;

        System.out.println("Enter elements:");

        for (int i = 0; i < r; i++) {
            for (int j = 0; j < c; j++) {
                a[i][j] = sc.nextInt();

                if (a[i][j] % 10 == 4) {
                    sum = sum + a[i][j];
                }
            }
        }

        System.out.println("Sum of elements ending with 4: " + sum);
    }
}

```

```

        }
    }
}

System.out.println("Sum = " + sum);
}
}

```

1(f). Print Upper Diagonal

```

import java.util.Scanner;

public class UpperDiagonal {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size of matrix: ");
        int n = sc.nextInt();

        int a[][] = new int[n][n];

        System.out.println("Enter elements:");

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Upper diagonal:");

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i <= j)
                    System.out.print(a[i][j] + " ");
                else
                    System.out.print("0 ");
            }
            System.out.println();
        }
    }
}

```

1(g). Print Lower Diagonal

```

import java.util.Scanner;

public class LowerDiagonal {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size of matrix: ");
        int n = sc.nextInt();
    }
}

```

```

int a[][] = new int[n][n];

System.out.println("Enter elements:");

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        a[i][j] = sc.nextInt();
    }
}

System.out.println("Lower diagonal:");

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        if (i >= j)
            System.out.print(a[i][j] + " ");
        else
            System.out.print("0 ");
    }
    System.out.println();
}
}

```

2(a). Accept Names and Sort Alphabetically

```

import java.util.Scanner;
import java.util.Arrays;

public class SortNames {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of names: ");
        int n = sc.nextInt();

        String names[] = new String[n];

        System.out.println("Enter names:");

        for (int i = 0; i < n; i++) {
            names[i] = sc.next();
        }

        Arrays.sort(names);

        System.out.println("Names in alphabetical order:");

        for (int i = 0; i < n; i++) {
            System.out.println(names[i]);
        }
    }
}

```

2(b). Count Total Number of Words and Characters

```
import java.util.Scanner;

public class WordsCharacters {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a sentence: ");
        String str = sc.nextLine();

        String words[] = str.trim().split("\\s+");

        System.out.println("Total words = " + words.length);
        System.out.println("Total characters = " + str.replace(" ",
"").length());
    }
}
```

3(a). Username Generator Using StringBuilder

```
import java.util.Scanner;

public class UsernameGenerator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter First Name: ");
        String first = sc.next();

        System.out.print("Enter Last Name: ");
        String last = sc.next();

        StringBuilder username = new StringBuilder();

        username.append(first.toLowerCase());
        username.append(".");
        username.append(last.toLowerCase());

        System.out.println("Generated Username: " + username);
    }
}
```

3(b). Check Whether Password is Strong

```
import java.util.Scanner;
```

```

public class StrongPassword {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        StringBuilder sb = new StringBuilder(password);

        boolean upper = false;
        boolean lower = false;
        boolean digit = false;
        boolean special = false;

        for (int i = 0; i < sb.length(); i++) {
            char ch = sb.charAt(i);

            if (Character.isUpperCase(ch))
                upper = true;
            else if (Character.isLowerCase(ch))
                lower = true;
            else if (Character.isDigit(ch))
                digit = true;
            else
                special = true;
        }

        if (sb.length() >= 8 && upper && lower && digit && special)
            System.out.println("Strong Password");
        else
            System.out.println("Weak Password");
    }
}

```

3(c). SMS Character Counter

```

import java.util.Scanner;

public class SMSCounter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter SMS: ");
        StringBuilder sms = new StringBuilder(sc.nextLine());

        int limit = 160;
        int used = sms.length();
        int left = limit - used;

        if (left < 0)
            left = 0;

        System.out.println("Characters Used : " + used);
        System.out.println("Characters Left : " + left);
    }
}

```

```
    }  
}
```

3(d). Shopping Bill Generator

```
import java.util.Scanner;  
  
public class ShoppingBill {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        StringBuilder bill = new StringBuilder();  
  
        System.out.print("Enter price of Milk: ");  
        int milk = sc.nextInt();  
  
        System.out.print("Enter price of Bread: ");  
        int bread = sc.nextInt();  
  
        System.out.print("Enter price of Butter: ");  
        int butter = sc.nextInt();  
  
        int total = milk + bread + butter;  
  
        bill.append("***** SHOP BILL *****\n");  
        bill.append("Milk      : ₹").append(milk).append("\n");  
        bill.append("Bread     : ₹").append(bread).append("\n");  
        bill.append("Butter    : ₹").append(butter).append("\n");  
        bill.append("-----\n");  
        bill.append("Total     : ₹").append(total);  
  
        System.out.println(bill);  
    }  
}
```

4. Reverse Each Word Using StringTokenizer

```
import java.util.Scanner;  
import java.util.StringTokenizer;  
  
public class ReverseWords {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter a sentence: ");  
        String sentence = sc.nextLine();  
  
        StringTokenizer st = new StringTokenizer(sentence);  
        StringBuilder result = new StringBuilder();  
  
        while (st.hasMoreTokens()) {  
            String word = st.nextToken();
```

```
        StringBuilder reverse = new StringBuilder(word);
        reverse.reverse();

        result.append(reverse).append(" ");
    }

    System.out.println("Output: " + result.toString().trim());
}
```

Example

Input: Java is fun to learn
Output: avaJ si nuf ot nrael